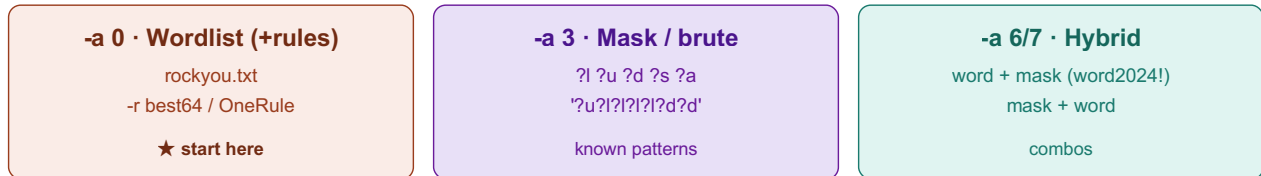


Hashcat

Crack captured hashes offline with GPU acceleration. Always: identify hash → pick mode (-m) → choose attack (wordlist / rules / mask) → run → --show. Companion to every capture cheatsheet (kerberoast, asrep, NTLM, NetNTLMv2). Replace <placeholders> with your own values.



Attack modes (-a)



Hash speed dictates strategy

FAST: MD5, NTLM, RC4 Kerberoast (\$23\$) → masks/brute feasible
SLOW: bcrypt, sha512crypt (\$6\$), AES tickets (\$18\$) → wordlist+rules only

Cracked passwords are cached in ~/.hashcat/hashcat.potfile and reused automatically.

No GPU? add --force (CPU, slow). John the Ripper auto-detects format as an alternative.

① Identify the Hash → Mode Number

Tab 1 · Kali

```
hashid '<hash>'
nth --text '<hash>'          # name-that-hash, suggests -m
```

Common modes (-m):

0	MD5	1000	NTLM	1800	sha512crypt (\$6\$)
100	SHA1	5600	NetNTLMv2	3200	bcrypt (\$2*\$)
13100	Kerberoast RC4 (\$krb5tgs\$23\$)	19700	Kerberoast AES256 (\$krb5tgs\$18\$)		
18200	AS-REP (\$krb5asrep\$23\$)	16500	JWT (HS256)	22000	WPA

② Wordlist (-a 0) — start here

Tab 2 · Crack

```
hashcat -m <mode> <hashfile> /usr/share/wordlists/rockyou.txt

# user:hash files (secretsdump / NTDS): add --username
hashcat -m 1000 ntlm.hashes /usr/share/wordlists/rockyou.txt --username
```

③ Wordlist + Rules (best value)

Tab 2 · Crack

```
hashcat -m <mode> <hashfile> rockyou.txt -r /usr/share/hashcat/rules/best64.rule
hashcat -m <mode> <hashfile> rockyou.txt -r /usr/share/hashcat/rules/
OneRuleToRuleThemAll.rule
```

best64 = fast; OneRuleToRuleThemAll = thorough. Rules mutate each word (digits, caps, leet) — huge coverage gain for weak-but-mutated passwords.

④ Mask / Brute (-a 3) & Hybrid (-a 6/7)

Tab 3 · Mask

```
# charsets: ?l lower ?u upper ?d digit ?s symbol ?a all
hashcat -m <mode> <hashfile> -a 3 '?u?l?l?l?l?l?d?d'          # Word12 style
hashcat -m <mode> <hashfile> -a 3 ?a?a?a?a?a?a --increment

# hybrid: wordlist + appended mask (word2024!)
hashcat -m <mode> <hashfile> -a 6 rockyou.txt '?d?d?d?d?s'
```

⑤ Run Control & Results

Tab 4 · Manage

```
hashcat -m <mode> <hashfile> <wordlist> -0 -w 3      # -0 optimized, -w workload
hashcat -m <mode> <hashfile> --show                  # print cracked (user:pass)
hashcat -m <mode> <hashfile> --left                  # still-uncracked
hashcat ... --session myrun                          # then --session myrun --restore
```

Live keys: 's' status, 'p' pause, 'q' quit. Estimate a mask's cost with --keyspace first. Kerberoast/asrep: prefer RC4 (\$23\$) — far faster than AES.

Key idea: cracking is offline and parallel — the only real levers are the hash type (some are deliberately slow: bcrypt, sha512crypt, AES tickets) and how smartly you attack (a good wordlist + rules beats blind brute force almost every time). Nail the mode number, throw rockyou + a rule at it first, then escalate to masks and hybrids only for what survives.